

Speed of Thought

Cost and performance are the parameters that determine whether a processor fits into the ‘budget’ solution category or is an out and out ‘performance’ solution. The prices of mainstream 32-bit processors have fallen substantially and you can now buy an Athlon XP 2400+ GHz processor for about Rs 5000 giving you performance equivalent of a Pentium 4 2.2 GHz which retails at around Rs 9,500.

Technology Insight

The processor is a device that receives information from software, processes it in a predetermined fashion and returns the results.

This information is sent to a small storage region known as Level 1 cache. The data cache stores data, and the instruction cache stores the instructions in the program that is currently running.

The Arithmetic and Logic Unit (ALU) forms the core of the processor and performs the actual mathematical processing on the data such as addition and subtraction. There are two ALUs, one which handles intergers and another that handles decimals or floating-point numbers, such as 75.13. The Floating Point Unit is responsible for handling any mathematical operations involving decimal numbers. The Pentium 4 is generally better at calculating floating-point numbers, and thus more suited for graphics and gaming. The AMDs have decent FPU performance but are much more cost effective, so they may be a good buy for running productivity applications.

The Branch Prediction Unit inspects the nature of the instructions in the instruction cache and decides which of the ALUs is best suited for processing the data fastest. This is done to ensure that no ALU is left idle. This streamlines the data transfer and thus increases the efficiency of the processor.

Decision Maker			
	General desktop use	Home entertainment	Gaming and graphics professionals
You need	Decent performance in office productivity Applications	Lag-free performance in all graphic, multimedia, and video oriented applications	Movies, music, gaming and complete home entertainment with oodles of processing power
Look for	Any Celeron/VIA class processor with speeds in excess of 1 GHz	A processor with speeds in excess of 900 MHz	An Athlon/Pentium 4 class processor with speeds in excess of 1.2 GHz
Our pick	Celeron 1.7 GHZ /AMD 1700+	Pentium 2.0 GHz or AMD 2400+	Pentium4 3.06 GHz or AMD XP 3000+
Price range	Up to Rs 5,000	Rs 8,000 to Rs 15,000	Rs 30,000 and above

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